MLOps: Experiment Tracking, Model Registry, and CI/CD for ML

Scaler School of Technology | ML System Design and LLMOps | Week 3 — Session 2/2 | Pre-Lecture Prep

What Is This Lecture About?

This session covers the operational side of ML: how to track experiments so you can reproduce them, how to manage model versions across environments, and how to build CI/CD pipelines that treat model training and evaluation as first-class engineering tasks. We also cover what happens after deployment—how to detect that your model is quietly degrading, and what to do about it.

Experiment Tracking and Reproducibility

Running hundreds of experiments without tracking them is like running hundreds of experiments without a lab notebook. You cannot reproduce your best result. You cannot explain why one model beat another. MLflow and Weights & Biases give you the lab notebook.

Model Registry and Lifecycle Management

A model file on S3 with a name like `model_v3_final_FINAL.pkl` is not a production artefact. A model in a registry has a semantic version, a lineage (what data, what code, what hyperparameters), lifecycle stages (Staging → Production → Archived), and a one-command rollback path.

CI/CD for ML and Distribution Shift

Software CI/CD tests code. ML CI/CD tests code *and* data *and* model performance. We will build a pipeline where new models are only promoted when they demonstrably beat the current production model. We will also cover the three types of distribution shift—covariate, label, and concept drift—and when to trigger retraining.

What to Expect in the Session

- Live MLflow demo: track an experiment end-to-end
- Model registry promotion walkthrough
- CI/CD pipeline diagram: code push → production
- Data validation gates with Great Expectations
- Distribution shift examples (COVID search queries)
- Retraining trigger design exercise
- Quiz: debug a degraded model in production

How to Prepare

- Think about your last ML project: could you reproduce your best result from scratch? Could you explain what changed between experiment 12 and experiment 13?
- Recall the concept of training-serving skew (Week 2): this session explains one of its root causes
- Optional: skim the MLflow quickstart at mlflow.org
- Optional: read Chip Huyen Ch. 8 opening story (grocery store prediction model that slowly went wrong)
- No new tools to install — we will use MLflow, which you likely already have via `pip install mlflow`

Duration: ∼2 hours **Prerequisites:** Weeks 1–3 Session 1 **Instructor:** Priyansh Saxena